



2.2 Prenatal Screening and Diagnosis

1. Now Screening: 'At Risk!' movie
 - prenatal screening tests → determine if fetus is at increased risk of a genetic d/o v.
 - prenatal diagnostic tests → determine if a genetic order is present
2. Movie on screen 1
 - first trimester screen (11-13wks GA, screens for T18/21) = PAPP-A, hCG (serum) + NT
 - (US, + if >3mm) = T21 detection rate 82-87%
3. PAPP Christmas on screen 1
 - first trimester screen (11-13wks GA, screens for T18/21) = PAPP-A, hCG (serum) + NT
 - (US, + if >3mm) = T21 detection rate 82-87%
4. Beta glasses on girl in santa's sleigh on screen 1
 - first trimester screen (11-13wks GA, screens for T18/21) = PAPP-A, hCG (serum) + NT
 - (US, + if >3mm) = T21 detection rate 82-87%

5

Translucent toy sack + 3 bullhorn ultrasound waves

5

first trimester screen (11-13wks GA, screens for T18/21) = PAPP-A, hCG (serum) + NT

(US, + if >3mm) = T21 detection rate 82-87%

6. Movie on screen 2

→ second trimester screen (15-22wks GA, screens for T18/21, NT + abd wall defects) =

“quad screen” = hCG, AFP, estriol, inhibin-A (serum) = T21 detection rate 81%

7. #18 train car falling off of the cliff

→ T18 + quad screen = ↓hCG, ↓AFP, ↓estriol, inhibin-A

8. Conductor w/ “21” tickets holding beta-glasses high + “A” stop light

→ T21 + quad screen = ↑hCG, ↓AFP, ↓estriol, ↑inhibin-A

9. Sign to “Downtown” + elevated “A” stop light

→ T21 + quad screen = ↑hCG, ↓AFP, ↓estriol, ↑inhibin-A

10. Tube shaped train car + ALF’s produce flying in the air

→ NT/abd wall defect + quad screen = hCG, ↑AFP, estriol, inhibin-A

11. “Integrated ticket \$0.96”

→ integrated screen = 1st tri (w/ NT) + 2nd tri “quad” screen = T21 detection rate 96%

12. “See ‘em ticket \$0.88”

→ serum integrated screen = 1st tri (withOUT NT) + 2nd tri “quad” screen = T21 detection rate 88%

13. “Sequential ticket with FREE DNA!”

→ sequential screening = 1st tri screen → (1) if normal → 2nd tri quad screen (2) if elevated → cfDNA or diagnostic testing

14. “Get your NIPBLES FREE!” snack stand

cell free DNA (AKA NIPS or NIPT) = can be done starting at 10wks GA, tests for T13/18/21 + fetal sex, T21 detection rate 99%

15. Older pregnant lady under “NIPBLES FREE!” sign

→ indications for cfDNA (AKA NIPS/NIPT) = abnormal screening test, abnormal anatomy
on US, AMA, prior pregnancy w/ fetal aneuploidy, known parental balanced translocation (ex. Robertsonian)

16. Abnormal ultrasound waves under “NIPBLES FREE!” sign

→ indications for cfDNA (AKA NIPS/NIPT) = abnormal screening test, abnormal anatomy
on US, AMA, prior pregnancy w/ fetal aneuploidy, known parental balanced translocation (ex. Robertsonian)

17. Chromosome twizzlers under “NIPBLES FREE!” sign

→ indications for cfDNA (AKA NIPS/NIPT) = abnormal screening test, abnormal anatomy
on US, AMA, prior pregnancy w/ fetal aneuploidy, known parental balanced translocation (ex. Robertsonian)

18. Robertson’s Snack Stand under “NIPBLES FREE!” sign

→ indications for cfDNA (AKA NIPS/NIPT) = abnormal screening test, abnormal anatomy
on US, AMA, prior pregnancy w/ fetal aneuploidy, known parental balanced translocation (ex. Robertsonian)

19. Drive-in menu with prices

→ T21 detections rates worst → best = second “quad” screen (81%), 1st tri alone (82-87%), serum integrated (88%), integrated (96%), cfDNA (99%)

20. Car with cracked windshield

→ common causes of abnormal prenatal screening other than genetic d/o = poor dating, multiple gestations

21. Couple fighting in cracked windshield car

→ common causes of abnormal prenatal screening other than genetic d/o = poor dating, multiple gestations

Quadruplets in cracked windshield car

22

common causes of abnormal prenatal screening other than genetic d/o = poor dating, multiple gestations

23. Tree trunk with spines

→ carrier screening = tests if a parent carries genes for any known diseases, usually AR, universally offer CF and SMA carrier testing

24. Tree with forked, sappy branches

→ carrier screening = tests if a parent carries genes for any known diseases, usually AR, universally offer CF and SMA carrier testing

25. Falling “H” in DRIVE IN T-EATRE sign

→ if anemia in a pregnant woman + high risk group (african, mediterranean, middle eastern, SE asian descent) → screen for hemoglobinopathies (sickle cell disease, alpha/beta-thalassemias) via hgb electrophoresis

26. “The Sickle” film reel

→ sickle cell disease = homozygous for HgbS (mutated beta-globin chain), electrophoresis findings = NO HgbA, ↑↑ HgbS

27. “The Sickle 2: Sickle Traitor” film reel

→ sickle cell trait = only 1 beta-globin gene has the sickle cell mutation, electrophoresis findings = ↓ HgbA, slight ↑ HgbS

28. α reel of “The Holy Chalice” film

→ α-thalassemia = impaired production of α-globin chains, # α chain genes = 4 = 4 possible diseases (0-1 normal α-chains → α-thal major)

29. β reel of “The Holy Chalice” film

→ β-thalassemia = impaired production of β-globin chains, # β-chain genes = 2 → 2 possible diseases. Hgb electrophoresis findings → HgbA2 (2α/2δ chains)

30

Star of David

30

if pregnant pt. is an Ashkenazi Jew → recommend screening for Tay-Sachs, Canavan

disease, familial dysautonomia, and cystic fibrosis

31. Sach of potatoes

→ if pregnant pt. is an Ashkenazi Jew → recommend screening for Tay-Sachs, Canavan disease, familial dysautonomia, and cystic fibrosis

32. “Can a van do this?” bumper sticker

→ if pregnant pt. is an Ashkenazi Jew → recommend screening for Tay-Sachs, Canavan disease, familial dysautonomia, and cystic fibrosis

33. Dent on back of car

→ if pregnant pt. is an Ashkenazi Jew → recommend screening for Tay-Sachs, Canavan disease, familial dysautonomia, and cystic fibrosis

34. Sappy branch hanging over car with Star of David

→ if pregnant pt. is an Ashkenazi Jew → recommend screening for Tay-Sachs, Canavan disease, familial dysautonomia, and cystic fibrosis

35. popCORN samples below screen 1

→ prenatal diagnostic testing = chorionic villus sampling (10-13wks GA) or amniocentesis (15-20wks GA) → samples sent for karyotype and/or microarray

36. Apple juice below screen 2

→ prenatal diagnostic testing = chorionic villus sampling (10-13wks GA) or amniocentesis (15-20wks GA) → samples sent for karyotype and/or microarray